Week 3 Worksheet 1

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Exercise 1. The hamiltonian for the harmonic oscillator is

$$H = \frac{1}{2m}[p^2 + (m\omega x)^2].$$

a) Show that

$$H = \hbar\omega(a^\dagger a + 1/2),$$

where

$$\begin{aligned} a^\dagger &= \alpha x + i\beta p \\ a &= \alpha x - i\beta p. \end{aligned}$$

That is, determine α and β.

b) Show that $[a, a^\dagger] = aa^\dagger - a^\dagger a = 1$.

c) Show that there is a solution ψ_0 to the TISE such that $a^\dagger a \psi_0 = 0$.
Hints: First, show that $[N, a] = -a$. If $N = a^\dagger a$ and $N\psi = \lambda\psi$ for some scalar λ and some normalizable ψ, show that $Na\psi = (\lambda - 1)a\psi$. Then, show that $|a\psi|^2 = \psi^* N\psi$. What does this tell you about repeated applications of a to ψ?

d) What is the energy of ψ_0?

e) Show that $N = a^\dagger a$ is the **number operator**: If $\psi_n = (a^\dagger)^n \psi_0$ is the n^{th} stationary state, then

$$N\psi_n = n\psi_n.$$

f) Show that if $N\psi_0 = 0$, then a complete set of solutions to the equation $N\psi = \lambda\psi$ is given by $(a^\dagger)^n \psi_0$, so that $(a^\dagger)^n \psi_0$ are also a complete set of solutions to the TISE.